

Literature Review: Definitions of Extreme Weather for Power System Forecast-Error Analysis

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Abstract

Background: No universal definition of extreme weather exists since appropriateness is purpose-dependent. In power-systems-and-forecasting research, “extremity” is increasingly framed in impact and forecast-error terms rather than meteorological rarity alone, but this practice is implicit and unsystematized. **Objective:** Evaluate definitions of extreme weather events that are most appropriate for analyzing the impact of forecast errors under extreme weather on power system operations compared to that under normal weather. **Eligibility:** Records combining weather extremity with forecasting or power-system operations and containing definitional, threshold, or identification content; infrastructure-resilience/restoration work without a definitional dimension was excluded. **Sources:** Scopus (n = 2,198; exported 2 July 2026). **Charting:** Each record tagged by definitional space (meteorological/impact/forecast-error) and construction mechanism. **Results:** A purposive top-50 core set was read in full. The literature is bifurcated: a majority defines extremity meteorologically but applies it operationally to segment forecasting regimes, while a smaller, more sophisticated strand defines extremity by operational consequence (loss-of-load) or forecast-error magnitude and explicitly critiques meteorological definitions. **Conclusions:** The main research will continue with meteorological thresholding due to the constraints of data availability and time but acknowledges this as an acceptable limitation given the constraints.

1. Introduction

1.1 Rationale

There is no universal definition of an extreme weather event; the appropriate definition depends on the analytical purpose. In the intersection of power-systems research and forecasting, “extreme” is increasingly operationalized through the impacts an event has on the system or the forecast errors it induces, rather than through meteorological rarity alone. This practice remains largely implicit, and no prior work has evaluated which definitional approach is most appropriate for forecast-error analysis specifically.

The motivating observation is that meteorologically ordinary events still drive operational stress. Chen et al. [1] show that individual-variable meteorological extremes—sustained low wind, low solar radiation, or extreme temperature—“do not always cause the most severe power shortages,” whereas “moderately extreme events that simultaneously reduce renewable supply and increase electricity demand can pose greater risks.” At equal occurrence frequencies, “these identified events pose several times the risk to power supply adequacy compared to individual-variable extremes and should be prioritized in power system planning” [1]. This review has a different aim than the recent taxonomy of Bromberger et al. [2], which classifies six weather-event types by physical, operational, and temporal attributes. That work taxonomizes impacts; this review evaluates definitions for a forecast-error purpose.

1.2 Objective

Our objective is to evaluate the definitions of extreme weather events most appropriate for analyzing the impact of forecast errors under extreme weather on power system operations compared to that under normal weather. Three candidate definitional spaces are considered: meteorological, impact-based, and forecast-error/predictability-based. In PCC form: Population = power-system operations research; Concept = definitions of extreme weather; Context = forecast-error analysis.

2. Methods

This scoping review follows the PRISMA-ScR reporting guidance [3].

2.1 Eligibility criteria

Included: records addressing weather extremity AND (forecasting OR power-system operations) with definitional, threshold, or identification content. Excluded: infrastructure-resilience, restoration, or recovery engineering with no definitional dimension.

2.2 Information sources

Scopus was the sole database (n = 2,198, exported 2 July 2026). All findings in §3, and references [1]–[18], derive from this corpus. References [19]–[27] are methodological sources hand-searched outside it, used only in §4; they do not enter the PRISMA-ScR flow and are excluded from every figure in §3.

2.3 Search strategy

A Scopus advanced query intersecting weather-extremity, definitional, and power-system term blocks, with a title-level exclusion of resilience/restoration/recovery (full string in Appendix). The mandatory power-system term places climatological and solar-radiation methodology outside the query's reach, hence the separate sourcing noted in §2.2.

2.4 Selection

Two-stage screening: 2,198 records → three-pillar intersection (definition × forecast × power) = 540 → strict operations filter = 354 → purposive relevance ranking (criteria: forecast-error orientation, definitional focus, operational impact, review status, recency/citation), yielding the top 50 as the read-in-full core set.

2.5 Data charting

Each record was tagged by definitional space (MET/IMP/FE) and construction mechanism — any rule contributing to whether an event counts as extreme (threshold, percentile, warning-criteria, index, extreme-value theory, ML/algorithmic, regime, net-load, ramp, persistence, spatial coherence). The axes are not fully independent: some mechanisms, such as net load and loss-of-load, are available only in impact space.

2.6 Data items

Definitional space, mechanism, variables, operational-impact-relevance, domain, year, citations.

2.7 Critical appraisal

Not conducted, consistent with scoping-review methodology.

2.8 Synthesis method

A two-axis taxonomy (space × mechanism) organizes the corpus. The comparative-evaluation template is modeled on the heat-wave-definition-comparison precedent in epidemiology—studies that rank alternative definitions against a health outcome, substituting operational outcomes (loss-of-load, forecast-error magnitude) for mortality.

3. Results

3.1 Selection

PRISMA-ScR flow: 2,198 identified → 540 after three-pillar intersection → 354 after strict operations filter → 50 in the read-in-full core.

3.2 Characteristics

The corpus is power-dominant (~74% power systems, ~6% health, ~11% agriculture) with a strong recency skew: roughly 57% of the core set is from 2025–2026, indicating an active, fast-moving research front.

3.3 Individual sources

Individual records are represented in the tagged top-50 table (charted per §2.5–2.6).

3.4 Synthesis

Among the top 50, the forecast-error space accounts for 26 records and the impact space 27 (categories overlap), while purely meteorological definitions survive as the primary space in only 1 record, indicating that meteorological variables persist chiefly as inputs to impact- or forecast-error-based definitions.

Hence, the literature is bifurcated. The majority defines extremity meteorologically and then applies it operationally, using meteorological thresholds to segment forecasting-model regimes. Liang et al. [4] exemplify the threshold mechanism, defining a “high-impact weather process” from two perspectives—abnormal output and abnormal prediction error—improving classification accuracy by 1.30% for a Jiangxi new-energy cluster [4]. Load-forecasting studies commonly segment regimes by quantile thresholds of temperature before fitting scenario-aware models. The wildfire load-forecasting model of Yang et al. [5] uses a categorical Fire Weather Index as the extremity descriptor, “significantly enhancing the average stability of forecast performances by 42%” and reducing MAPE by roughly 14.37% (Bi-GRU) and 20.86% (Vision Transformer). Crucially, it demonstrates that “load forecasting during wildfire seasons is more challenging than during non-wildfire seasons,” with “the improvement rate of load forecasting performance during wildfire seasons[...] more than two times greater than in non-wildfire seasons” [5], which is a direct forecast-error-differential result of the kind the main research targets.

A smaller, more sophisticated strand defines extremity by operational consequence or forecast-error magnitude, and explicitly critiques meteorological definitions. Chen et al. [1] define extremity through a rapid loss-of-load estimation (computed “hundreds of times faster than dispatch optimization models”), identifying events that “often involve moderately low VRE output and moderately high demand occurring simultaneously, resulting in severe shortages.” Hu et al. [6] identify stress events as 17 or 18 consecutive hours of extremely high residual load, finding these short-duration events “can impose significant stress on the energy system, often exceeding the

impact of longer-term extreme events” [6]. In the European dunkelflaute study [7], extremity is defined by expected energy not served (ENS) during a worst-case two-week low-renewable period across three 2035 electricity-mix scenarios in 11 North-Western European countries. “During the dunkelflaute period, ENS is observed in all scenarios, with Germany particularly at risk, experiencing an hourly average ENS at GWh scale” [7]. The PJM offshore-wind study [8] anchors extremity in loss-of-load expectation (LOLE), reporting that hurricane-driven outages raise LOLE modestly under typical conditions (0.22 hours/year increase) but up to 3.54 hours/year under combined extreme-weather and high-electrification scenarios [8]. The HVDC-flexibility study explicitly places “excessive resource forecast errors” alongside heatwaves as “extreme operating conditions,” finding HVDC flexibility avoids 24.8–28% of unserved energy under such conditions [9].

Mechanism distribution. Across the on-scope set ($n = 930$), ML/algorithmic detection is the largest mechanism ($\approx 24\%$; 23 of the top 50), spanning clustering, CNN/LSTM, transformers and GANs — for example, cold-wave discrimination for wind-farm-cluster forecasting [10]. Its prominence reflects the corpus’s 2025–2026 skew: an active front, not a settled convention. Impact/adequacy mechanisms follow ($\approx 22\%$), then fixed absolute thresholds ($\approx 16\%$; e.g. 35°C , 25 m/s), persistence criteria ($\approx 13\%$, almost always layered on another mechanism), extreme-value theory ($\approx 7\%$) and agency warning criteria ($\approx 6\%$, calibrated in this corpus almost exclusively for public health rather than grid stress). Percentile cutoffs are comparatively modest ($\approx 4\%$), as are composite indices such as WBGT and the Fire Weather Index ($\approx 2\%$) [5].

The impact/adequacy cluster is broad in gesture but narrow in rigor. Most of its 22% merely invokes outages or curtailment as consequences; requiring a formal metric — LOLE, EENS, resource adequacy — collapses it to $\approx 2\%$, and driver-side measures (net load, residual load) to $\approx 1\%$. Both sit in impact space, divided by position on the causal chain: outcome-side measures quantify the shortfall once the system fails to cope [1,8,12], driver-side measures the stress producing it [6,7] — the latter more tractable, being computable from weather and load without a dispatch simulation. Ramp/rate-of-change [11], which thresholds the derivative rather than the level, and spatial-coherence mechanisms likewise remain minority choices. The mechanisms best matched to grid stress are precisely those the literature underuses.

Operational metrics appear as outcomes, not cutoffs. Notably, LOLE (hours/year), EENS percentages and MAPE/RMSE are reported as outcome measures rather than as definitional thresholds. The gas-load study [13] reports its error rising +56.7% under extreme weather against a competitor’s +109.2%; the capacity-credit study [14] validates its method against cold-wave events; and the storage-allocation [15] and tunnel-microgrid [16] studies report loss-of-load and EENS reductions as consequences of design interventions. Operational definitions are therefore inherently system-relative rather than universal: the same MAPE or LOLE value means different things in different systems.

4. Proposed Meteorological Threshold Method

This section specifies the meteorological thresholding method for the main research. The rationale for adopting it over the theoretically preferred operational definition is given in §5.1–5.3.

4.1 Rationale and design principles

A *relative* (percentile) cutoff is preferred over a *fixed absolute* one: it applies a single rule across heterogeneous variables, transfers across locations without site-specific calibration, and handles two-tailed variables symmetrically [19]. Four components are applied in sequence to each variable: variable selection → deseasonalization → percentile threshold → persistence criterion. Extremity is defined on *observed* meteorology and forecast error evaluated *within* the resulting regimes (§4.6), keeping the independent variable (was the weather extreme?) distinct from the dependent one (how large was the error?).

4.2 Variable selection

Eight variables are available: GHI, temperature (T), relative humidity (RH), surface pressure (p), wind speed, the components U10 and V10, and wind direction. They are not equally suited to scalar thresholding, and the set is reduced as follows.

Thresholded. GHI, temperature, and wind speed: each has a clear extreme ordering and a direct link to power-system stress. Wind speed is derived from the components:

$$v = \sqrt{U_{10}^2 + V_{10}^2}$$

Reserved. RH and pressure are not thresholded alone, for three reasons: pressure’s power-system effect is indirect and already captured through the variables it drives (a deep low appears as a wind extreme, a blocking high as a heat or low-wind extreme); RH is impactful only in conjunction with heat or cold rather than alone; and both are poorly suited to level-based percentile cutoffs (RH saturates at 100%, pressure extremes are better described by rate-of-change). RH is therefore retained only as a compound ingredient and pressure only as a synoptic-regime indicator.

Excluded. U10 and V10 are directional decompositions, so an “extreme U10” is meaningless without direction; they serve only to construct v . Wind direction is circular, with no magnitude ordering, so it is not thresholdable as a scalar and is retained only for conditioning.

4.3 Deseasonalization and baseline

Percentiles must be computed against a baseline that removes the strong diurnal and seasonal non-stationarity of the variables. Without it, a percentile mislabels ordinary states as extreme — every afternoon a “temperature extreme,” every night a “low-GHI extreme.”

Baseline dataset and period. Thresholds are computed from ERA5 reanalysis over the 1991–2020 period — the current 30-year climatological standard normal recommended by WMO [20]. ERA5 (hourly, 1940–present) supplies the long record that the HRRR study data (2022–2024) cannot, giving stable percentiles even after slicing by hour-of-day. The baseline period ends before the 2022–2024 study period, so the reference climate is independent of the data being classified; the modest warming between the baseline midpoint and the study period is noted in §5.2.

Removing the seasonal cycle. The mean seasonal cycle is subtracted from the record before any percentile is computed. Brunner and Voigt [25] show that applying a running window directly to raw data mixes the mean

seasonal cycle into the extreme threshold, biasing the exceedance frequency by as much as -75% in regions with a strong seasonal gradient and weak day-to-day variability — among them the western United States. Removing the mean cycle first eliminates this bias and, in their words, allows longer windows to be used without penalty. The cycle is the empirical mean for each day-of-year d and hour-of-day h , averaged over the N_y baseline years:

$$\bar{x}(d, h) = \frac{1}{N_y} \sum_{y=1}^{N_y} x(y, d, h)$$

and the deseasonalized anomaly is the observation minus this cycle:

$$x'(t) = x(t) - \bar{x}(d(t), h(t))$$

Moving-window percentiles. Thresholds are then computed on the anomaly series. Following the calendar-day percentile methodology [19] with the correction of [25], the threshold for a given d and h is the p -th percentile of the anomalies falling in a two-dimensional window — ± 15 days in the calendar direction (a 31-day window) and ± 1 hour in the diurnal direction — pooled across all baseline years:

$$Q_p(d, h) = \text{percentile}_p \{ x'(\tau) : \tau \in \mathcal{W}(d, h) \}$$

The window is indexed by hour-of-day so that the diurnal cycle is not pooled away, and the 31-day calendar width is the configuration validated by [25]: once the mean cycle has been removed, the wider window simply increases the sample behind each threshold (on the order of $30 \text{ years} \times 31 \text{ days} \times 3 \text{ hours} \approx 2,800$ anomalies) without reintroducing seasonal bias. Only the mean cycle is removed, not its seasonal variance; the adequacy of this mean-only correction is treated as a sensitivity question (§4.7).

ERA5-to-HRRR transfer. Because thresholds are derived from ERA5 (≈ 31 km) but applied to HRRR (≈ 3 km), the two distributions can differ, most notably for wind in complex terrain. Thresholds are therefore bias-corrected to the HRRR climatology per site using the 2022–2024 overlap before application; the size of this correction is reported per variable as a diagnostic, and its residual uncertainty is noted in §5.2.

GHI. Dominated by deterministic solar geometry, GHI is replaced by the clear-sky index — the ratio of observed to modeled clear-sky irradiance [21,22] — with night-time hours excluded:

$$k_t(t) = \frac{\text{GHI}(t)}{\text{GHI}_{\text{clear-sky}}(t)}$$

This isolates cloud- and aerosol-driven variability, the component relevant to solar shortfalls, from the predictable diurnal cycle. Because night-time exclusion shrinks the effective sample, the observation count behind each GHI threshold is reported as a diagnostic.

4.4 Percentile threshold and per-variable direction

An observation is flagged as extreme when its anomaly falls beyond the window percentile threshold. For an upper-tail extreme:

$$E_{\text{high}}(t) = \mathbb{1}[x'(t) \geq Q_{90}(d(t), b(t))]$$

and for a lower-tail extreme:

$$E_{\text{low}}(t) = \mathbb{1}[x'(t) \leq Q_{10}(d(t), b(t))]$$

where x' is the deseasonalized anomaly, Q_{90} and Q_{10} are the window percentiles of §4.3, and $\mathbb{1}[\cdot]$ is the indicator function.

Percentile levels. The primary thresholds are the 90th and 10th percentiles. These inclusive levels are deliberate: comparing error distributions between regimes requires enough extreme-condition samples for stable estimation, and stricter levels (95th/99th) yield purer extremes but too few cases. The same trade-off is documented in the heat-wave literature, where the 90th percentile is selected as the balance between events being sufficiently “extreme” and sufficiently “measurable,” since higher percentiles combined with a persistence requirement leave too few events to analyze [23]. The 95th/5th level is retained as a robustness check (§4.7).

Relevant tail by variable.

Variable	Tail(s) of interest	Operational rationale
Temperature	Both ($\geq 90\text{th}$ and $\leq 10\text{th}$)	Heat and cold both drive load extremes and affect generation
Wind speed	Primarily low ($\leq 10\text{th}$); optionally high ($\geq 90\text{th}$)	Low wind $\rightarrow$ generation drought / adequacy risk; high wind $\rightarrow$ turbine cut-out
GHI (as k_t)	Low ($\leq 10\text{th}$)	Low irradiance $\rightarrow$ solar-generation shortfall

Where both tails of a variable are used, extreme status is the union of the high and low indicators.

4.5 Persistence (duration) criterion

Isolated single-timestep exceedances are not treated as events. An extreme event requires the threshold to be met for at least D consecutive time steps:

$$\text{Event}(t) = \bigwedge_{k=0}^{D-1} E(t - k)$$

Adapting the “wave” convention (≥ 3 consecutive days for daily data) to hourly resolution, the default is $D = 6$ consecutive hours, tested across 3–24 hours. Below three hours, exceedances reflect transient fluctuation rather than a sustained regime; six hours captures a genuine condition (a ramp, a lull, a heat build-up) while retaining the

short compound events the loss-of-load literature flags as high-impact; the upper range tests whether the contrast strengthens for longer events.

4.6 Regime partitioning and the normal reference class

The extreme indicator partitions the record into two regimes: **extreme** (hours in a qualifying event per §4.4–4.5) and **normal**. Defining the normal reference class carefully matters, because a poor choice confounds the extreme/normal contrast with season or background regime. Two considerations from the literature are combined here. First, following the percentile-complement approach used in machine-learning forecast evaluation against a deseasonalized climatology [26], a normal hour must have an anomaly in the central body of the 30-year ERA5 distribution — strictly between the two percentile thresholds, optionally tightened to the 40th–60th band — so that near-threshold hours do not contaminate the reference class. Second, following the matched-control design of heat-wave impact studies, which compare event periods against the days immediately before and after each event [27], a normal hour must also fall within ± 5 days of the boundaries of a qualifying extreme event. Third, the persistence criterion of §4.5 is applied symmetrically: a normal hour must belong to a run of at least D consecutive hours that are themselves climatologically central and non-extreme. Without this, the two regimes would differ in stability as well as extremity — extreme hours would all sit within sustained regimes while normal hours could be isolated moments between near-threshold conditions — and any error difference attributable to regime transience could not be separated from the effect of extremity. Because the central body of the distribution is where the atmosphere spends most of its time, this requirement excludes comparatively few candidate hours. A normal-reference hour therefore satisfies all three conditions — climatologically central, temporally local to an event, and part of a sustained run of length D — and is never itself part of any event. The temporal matching holds season and slow-varying background conditions approximately fixed, so the forecast-error comparison isolates the effect of extremity rather than of the calendar. The combination of a climatological-normalcy screen with event-local matching, and the symmetric application of D , are adopted here as design choices rather than inherited from a single source; the ± 5 -day window follows the pre/post-event bracket of [27] and is varied in the sensitivity analysis (§4.7).

Forecast error is then computed *within* each regime. Because error grows with horizon regardless of weather, pooling all 48 lead times would confound that growth with the extreme-versus-normal contrast. Metrics are therefore computed per lead time h , or in bands (1–6, 7–24, 25–48 h), and compared across regimes within each band:

$$\text{RMSE}_{\text{regime}}(h) = \sqrt{\frac{1}{|\mathcal{T}_{\text{regime}}|} \sum_{t \in \mathcal{T}_{\text{regime}}} (\hat{y}(t+h | t) - y(t+h))^2}$$

with bias, MAE, and MAPE evaluated analogously. The quantity of interest is the *difference* in error distributions between regimes at matched lead time. Because extremity is defined on observations, the forecast never enters the partition; it is evaluated only after the partition is fixed. A dual-threshold design — defining extremity jointly on observations and forecasts to isolate forecast “busts” — is deferred to future work.

4.7 Parameter choices and sensitivity

The method has five free parameters, each with a stated default and a tested range. The baseline period is fixed to the WMO 1991–2020 normal (§4.3) and is not varied.

Parameter	Default	Sensitivity range
Percentile level	90th / 10th	95th / 5th
Calendar window	31 days (± 15)	15–31 days
Diurnal pooling	± 1 hour	0 / ± 2 hours
Seasonal correction	mean removed	mean + variance
Normal matching window	± 5 days	± 3 – ± 7 days
Persistence D (both regimes)	6 hours	3–24 hours
Lead-time resolution	bands: 1–6 / 7–24 / 25–48 h	per-hour (1–48 h)
Normal-band definition	between the two tails	40th–60th core band

The persistence parameter D is varied for both regimes together (§4.6), preserving their comparability at every point in the sweep. Reporting results across these ranges shows the extreme/normal error contrast to be a property of the data rather than an artifact of one parameterization — the minimum robustness standard for a threshold-based definition. This practice follows sensitivity studies of extreme-weather definitions, which find that the percentile and duration choices materially change the measured relationship to a downstream outcome and that the most appropriate combination is outcome-dependent rather than universal, so that no single definition can be assumed optimal a priori [23,24]. A harmonic (Fourier) deseasonalization is retained as an alternative baseline construction: agreement between the two confirms the flagged events are not an artifact of the windowing choice.

5. Discussion

5.1 Summary of evidence

Meteorological thresholding remains the feasible base layer for candidate-event identification—it is data-available, transferable across systems, and computationally cheap. But impact- and forecast-error-based criteria determine actual appropriateness: they capture the compound and moderate events that meteorological rarity misses [1], and they align directly with the operational quantities a forecaster cares about. The leading literature already operates this way implicitly but has not standardized the practice. The method proposed in §4 reflects this reading: it retains meteorological thresholds as the identification layer but constructs them on established climatological foundations — seasonal-cycle removal [25] followed by calendar-day moving-window percentiles against the WMO 1991–2020 normal [19,20] — while deferring the appropriateness question to a forecast-error comparison in which extremity is fixed on observations and error is resolved by lead time (§4.6).

5.2 Limitations

Given constraints on data availability and the project timeline, this review adopts meteorological thresholding as the working definition of extremity. This is a known trade-off: the operational-definition literature shows that

meteorological thresholds can miss moderately extreme, compound events that pose greater shortfall risk than any single-variable extreme [1]; this is carried forward as an accepted scope constraint. Two data-related limitations attach to the §4 method. First, the study data (HRRR, 2022–2024) is short, and although the ERA5 1991–2020 baseline supplies a stable climatology, the baseline predates the study period, so extreme-heat events are defined against a marginally cooler climate. Second, thresholds are transferred from ERA5 (≈ 31 km) to HRRR (≈ 3 km); the bias-correction step (§4.3) mitigates but does not eliminate the resolution mismatch, which is largest for wind in complex terrain. Third, the seasonal-cycle removal of §4.3 subtracts only the mean cycle, following [25]; the seasonal variation in spread is left in the anomalies, so the mean-only correction is tested against a mean-plus-variance alternative in the sensitivity analysis (§4.7). Fourth, the event-local matching of the normal reference class (§4.6) reduces the number of normal-regime hours; the ± 5 -day window is widened in the sensitivity analysis where the normal sample would otherwise be too small for stable error estimation at a given lead time. Driver-side operational mechanisms (net load, residual load) are underdeveloped in the retrieved corpus, so conclusions about them rest on a thin evidence base. The strong recency/ML skew means older foundational definitional work was largely filtered out by the ranking system. The purposive top-50 selection was optimized for the review objective and is disclosed transparently rather than claimed as exhaustive. References [19]–[27] fall outside the reproducible search strategy; they are kept separate from [1]–[18] so that no claim about the definitional landscape rests on them. Keyword-derived tags were verified by close reading of abstracts and full texts where retrievable.

5.3 Conclusions

Ideally, we would select the operational definition of extremity as the approach for the main research going forward. However, due to constraints on data availability and the project timeline, we have chosen meteorological thresholding as the working definition of extremity for the main research, rather than the impact/forecast-error-based definition identified as most theoretically appropriate in §5.1. This choice has precedent: threshold-based extreme-event identification is used specifically to cope with rare-event data scarcity in short-term wind power forecasting under limited samples [17], and is similarly adopted for computational efficiency in climate-data-constrained grid-planning frameworks [18]. Meteorological thresholds are data-available, reproducible, and tractable without requiring a large historical record of loss-of-load or shortfall events. Concretely, the main research will apply the percentile method of §4: the mean seasonal cycle is removed [25], then 90th/10th moving-window percentiles are computed on the anomalies over the WMO 1991–2020 ERA5 normal [19,20] with a 31-day window and bias-corrected to HRRR, with irradiance handled through a clear-sky-index transform [21,22] and forecast error compared between observation-defined extreme and normal regimes at matched lead time. These properties make the approach the practical choice for this phase of the research, even though it is not the definition this review’s synthesis found most operationally appropriate.

6. Appendix: Scopus advanced search query

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( TITLE-ABS ( "extreme heat" OR "heat wave*" OR "heatwave*" OR "cold spell*" OR "cold wave*" OR
"extreme temperature*" OR "extreme wind*" OR "wind drought" OR "renewable drought" OR "dunkelflaute"
OR "compound event*" OR "extreme weather" OR "weather extreme*" OR "severe weather" OR "anomalous
weather" ) ) AND ( TITLE-ABS ( "defin*" OR "threshold*" OR "percentile*" OR "criteri*" OR "identif*"
OR "detect*" OR "classif*" OR "index" OR "indices" OR "metric*" OR "characteriz*" OR "framework" ) )
AND ( TITLE-ABS-KEY ( "power system*" OR "electricity" OR "electric grid" OR "power grid" OR "energy
system*" OR "renewable energy" OR "wind power" OR "solar power" OR "photovoltaic*" OR "net load" OR
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"residual load" OR "net demand" OR "load forecast*" OR "power forecast*" OR "wind forecast*" OR "solar forecast*" OR "forecast error*" OR "prediction error*" OR "supply adequacy" OR "resource adequacy" OR "loss of load")) AND NOT (TITLE ("resilience" OR "restoration" OR "recovery")) AND (LIMIT-TO (LANGUAGE , "English"))

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